- Instance Characteristics
- EC2 Purchase Options
- RI Flexibility Factor
- RI Coverage
- Saving Plans Coverage
- Spot Risks
- Spot Recommendations

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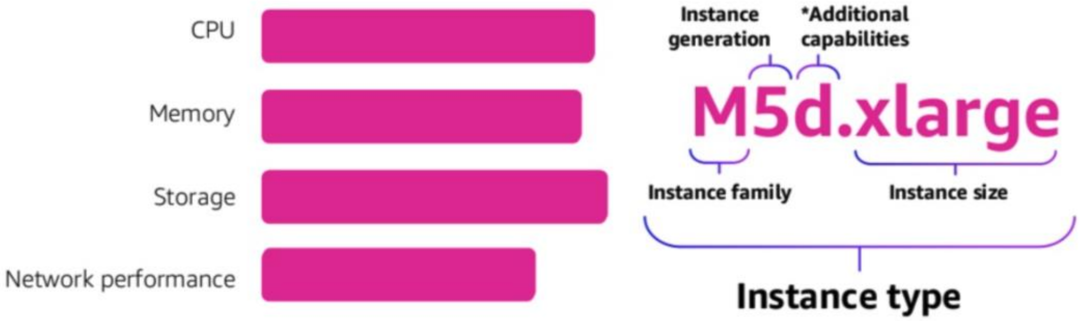
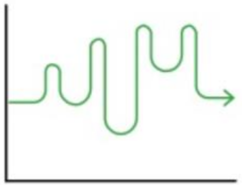
AWS EC2 Instance Characteristics



AWS EC2 Purchase Options

On-Demand

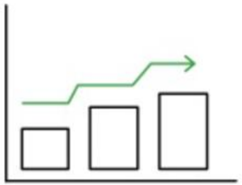
Pay for compute capacity by **the second** with no long-term commitments



Spiky workloads, to define needs

Reserved Instances

Make a 1- or 3-year commitment and receive a **significant discount** off On-Demand prices



Committed and steady-state usage

Spot Instances

Spare Amazon EC2 capacity at **savings of up to 90%** off On-Demand prices

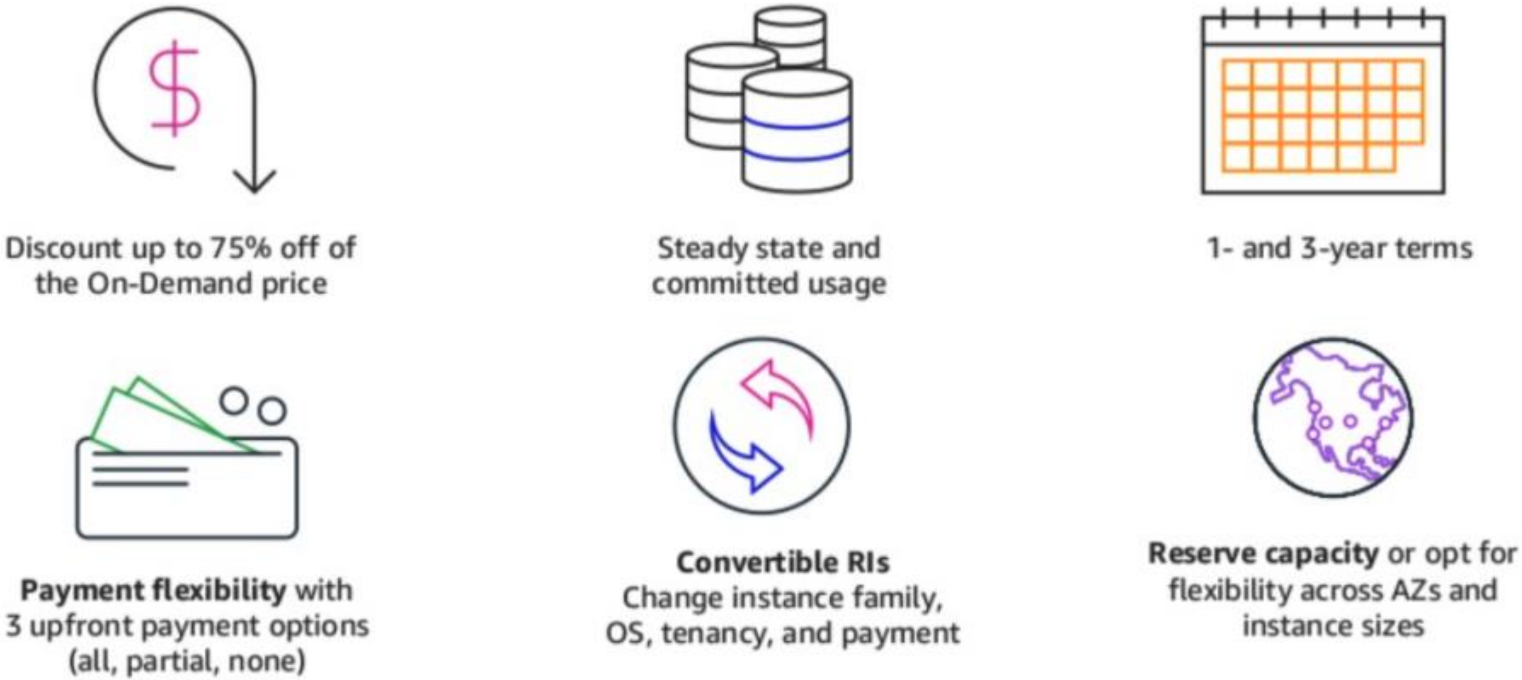
Fault-tolerant, flexible, stateless workloads

Dedicated Instances – Instance (hr) running on single-tenant hosts (account isolation).

Dedicated Hosts – Physical host (hr) fully dedicated for you (hardware isolation).

Capacity Reservations – Reserve capacity for EC2 in a specific AZ for any duration.

AWS EC2 Reserved Instances



AWS EC2 RI vs. Savings Plans

Reserved Instances

- EC2, RDS, ElastiCache, ElasticSearch, RedShift
- Specific instance family/size/region(/az) (e.g. r5, db.r5)

Saving Plans

EC2

- EC2 specify family (e.g. r5, db.r5)
- Any size/region/tenancy/os

Compute

- EC2, Fargate and Lambda
- Any family/type/region/tenancy/os

SageMaker

https://console.aws.amazon.com/cost-management/home#/dashboard

AWS EC2 RI - Flexibility Factor

Instance Size	Normalization Factor
nano	0.25
micro	0.5
small	1
medium	2
large	4
xlarge	8
2xlarge	16
4xlarge	32
8xlarge	64
10xlarge	80
16xlarge	128
32xlarge	256

It means that:

- 1 large = 4 small
- 1 large = 8 nano
- 1 xlarge = 8 small
- 1 2xlarge = 2 xlarge
- And so on...

https://aws.amazon.com/blogs/aws/new-instance-size-flexibility-for-ec2-reserved-instances/

AWS EC2 RI - Visibility and Coverage

Root Account ⇔ Payment Account

Root Account

RDS RI:
1x db.t3.xlarge
9 x db.r5.xlarge

1. Reservations are only visible (Cost Explorer) and applicable (Coverage) in purchasing account!

2. Recommendations, Purchase Granularity, Avoid Fragmentations

Member Accounts

A Account … B Account … C Account

RDS RI:
1 x db.t3.small

RDS:
db.t3.small

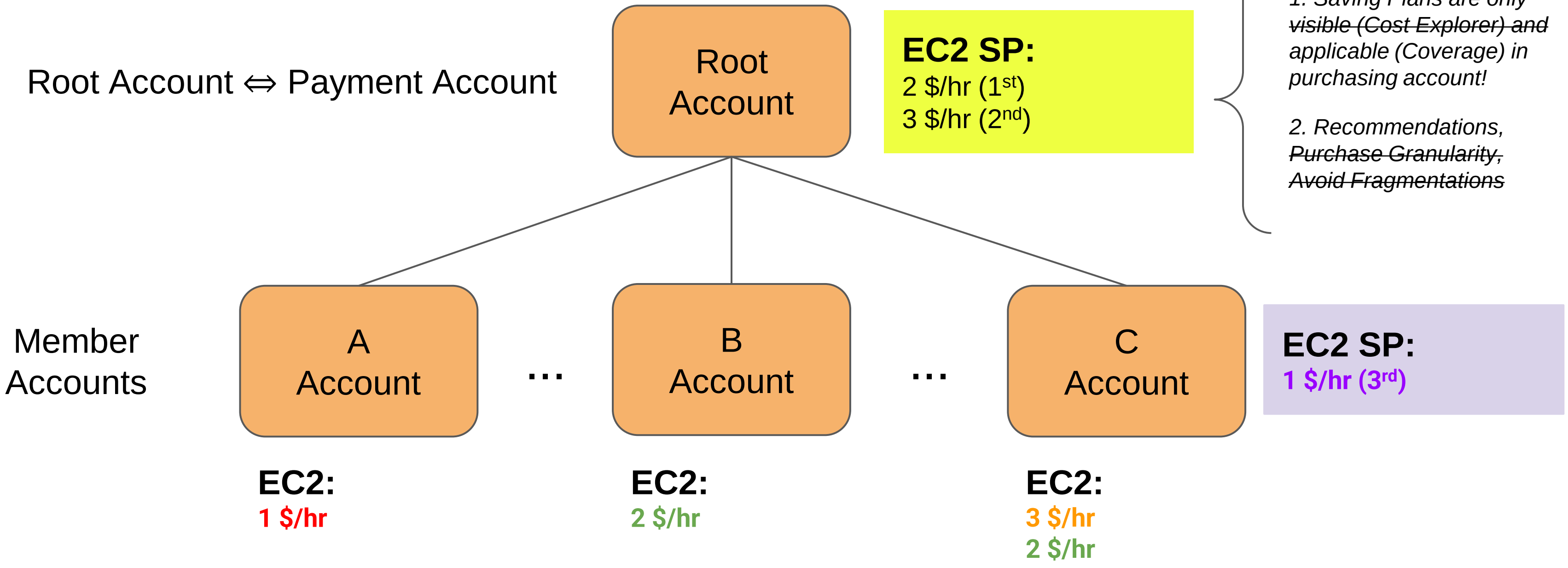
RDS:
db.t3.xlarge

RDS:
db.r5.2xlarge
db.r5.8xlarge

Fully Covered / Partially Covered / Not Covered! / Unused!

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-reserved-instances.html

AWS EC2 Saving Plans - Visibility and Coverage



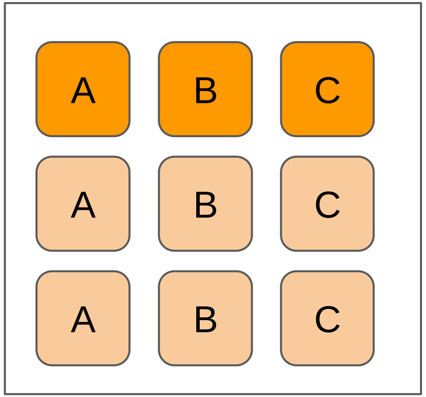
1. Saving Plans are only ~~visible (Cost Explorer) and~~ applicable (Coverage) in purchasing account!

2. Recommendations, ~~Purchase Granularity, Avoid Fragmentations~~

Fully Covered / Partially Covered / Not Covered! / Unused!

https://docs.aws.amazon.com/savingsplans/latest/userguide/sp-applying.html

AWS EC2 Spot Risk - Scenario A

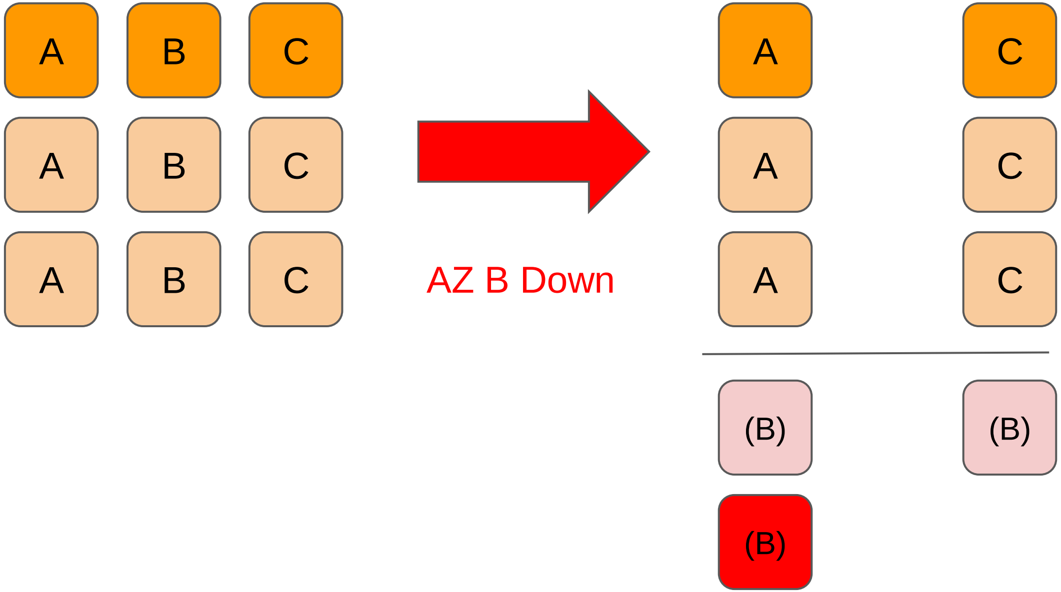


30% On-Demand
70% Spot

- Daily/Stable
- Spikes on Spot

AWS EC2 Spot Risk - Event A

30% On-Demand
70% Spot



22% On-Demand
44% Spot

66% Alive

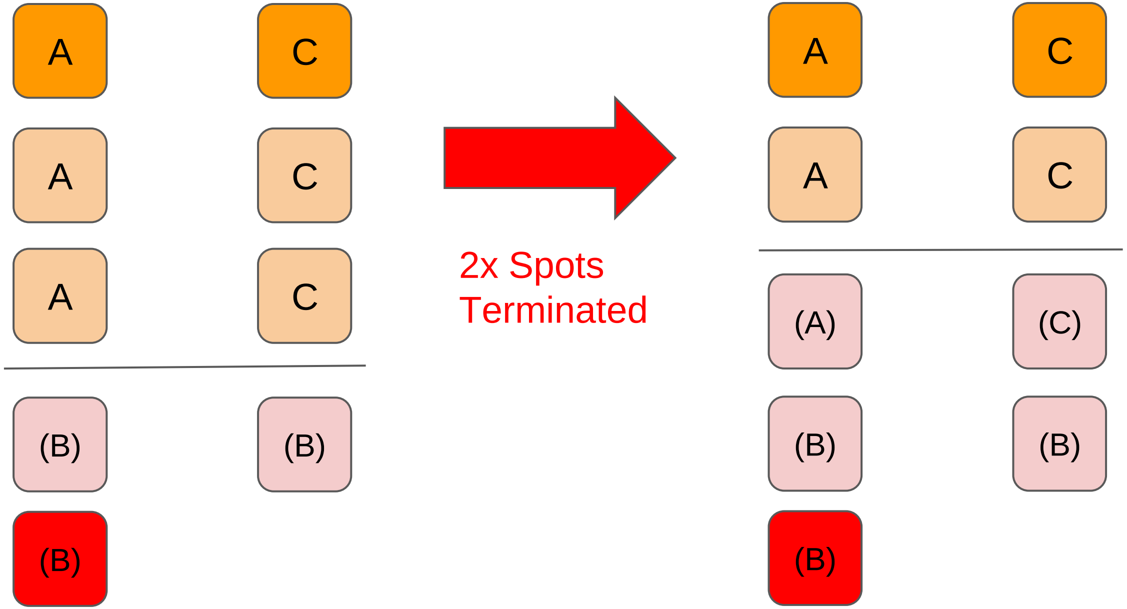
~ 34% Unknown

AWS Service Priority:
On-Demand > Spot

AWS EC2 Spot Risk - Event B

22% On-Demand
44% Spot

~ 30% Unknown



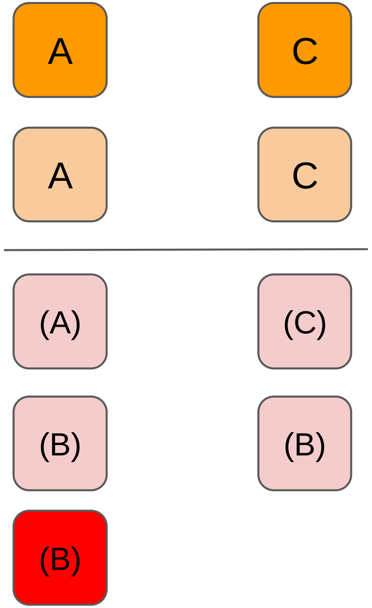
22% On-Demand
22% Spot

44% Alive

~ 66% Unknown

AWS Service Priority:
On-Demand > Spot

AWS EC2 Spot Risk - Same Outcome...



22% On-Demand
22% Spot

44% Alive

~ 66% Unknown

Events: A=>B ⇔ B=>A ⇒ ~66% Unknown

AWS EC2 Spot Recommendations

- **Production:**
  - **User facing applications:** >= 60% on-demand
  - **Other:** >= 30% on-demand
- **Non-Production:**
  - **Any:** 100 % spot
- **All**, be prepared for:
  - Test killing all your EC2 Spot in one AZ.
  - Test killing/rebooting all your EC2 Spot/On-Demand in one AZ.

AWS EC2 Spot When to Use Them

- You can use Spot Instances for various fault-tolerant and flexible applications. Examples include stateless web servers, API endpoints, big data and analytics applications, containerized workloads, CI/CD high performance and high throughput computing (HPC/HTC), rendering workloads, and other flexible workloads.
- Spot Instances are not suitable for workloads that are inflexible, stateful, fault-intolerant, or tightly coupled between instance nodes. **Spot Instances are also not recommended for workloads that are intolerant of occasional periods when the target capacity is not completely available. We strongly warn against using Spot Instances for these workloads or for attempting to fail-over to On-Demand Instances to handle interruptions.**

https://docs.aws.amazon.com/whitepapers/latest/cost-optimization-leveraging-ec2-spot-instances/when-to-use-spot-instances.html

AWS EC2 Lifecycle

To determine the instance lifecycle using the AWS CLI

Use the following describe-instances command:

```
aws ec2 describe-instances --instance-ids i-1234567890abcdef0
```

If the instance is running on a Dedicated Host, the output contains the following information:

```
"Tenancy": "host"
```

If the instance is a Dedicated Instance, the output contains the following information:

```
"Tenancy": "dedicated"
```

If the instance is a Spot Instance, the output contains the following information:

```
"InstanceLifecycle": "spot"
```

Otherwise, the output does not contain `InstanceLifecycle`.

AWS EC2 Lifecycle (Kubernetes)

```
kubectl get nodes -o custom-
columns=NAME:.metadata.name,ACC:.metadata.labels.application,TYPE:.metadata.labels.node_type,LIFECYCLE:".metadata.labels.aws\.io/li
fecycle" --sort-by .metadata.labels.application
```

Lifecycle Normal ⇔ On-Demand